
THE BIOMECHANICAL AND PERCEPTUAL INFLUENCE OF CHAIN RESISTANCE ON THE PERFORMANCE OF THE OLYMPIC CLEAN

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ABSTRACT

Proponents of chain training suggest that using chains hung from the ends of barbells rather than using conventional barbells alone enhances strength, power, and neuromuscular adaptations. The purpose of this study was to determine whether a conventional barbell with chains compared to a conventional barbell without chains would affect the performance of an Olympic Clean. The subjects were also asked regarding their perception of how chains affected their lifting. Four male and 3 female competitive weightlifters who used chains as part of their training participated in the study. The testing protocol compared the subjects' lifting 80% and 85% of their 1 repetition maximum (1RM) using conventional barbells and their lifting 80% and 85% of their 1RM using chains (75% conventional barbells + 5% chains and 80% conventional barbells + 5% chains, respectively). Video analysis evaluated the bar's vertical displacement and velocity and the rate of force production. Vertical ground reaction forces for the first-pull, unweighting, and second-pull phases of the lift were evaluated by using a force plate. After testing, the subjects completed a 2-item questionnaire asking individual perception of the effects of the chains. The results showed no significant difference for condition for any of the variables examined. In contrast, all subjects perceived that the chains required a greater effort. In conclusion, the results indicated that the addition of chains provided no greater value over lifting conventional barbells alone in the performance of the Olympic Clean, although the subjects perceived the chains to have a positive effect.

KEY WORDS chain training, weightlifting, uncommon implements

INTRODUCTION

The unconventional training method of attaching chains to the ends of bars has become common in commercial gyms, high schools, and collegiate strength and conditioning programs (1,2,4,6,10,11,19,20). Chain training requires chains to be hung from the ends of conventional barbells and to drape to the floor. The conventional barbell with attached chains then is used during exercises such as the squat, bench press, deadlift, and Olympic lifts (4,6). Theoretically, chains provide a variable resistance unlike conventional barbells alone. When lifting a conventional barbell alone, the weight on the barbell (i.e., external resistance) remains fixed. Although the external resistance remains constant, the force exerted by the working muscles varies throughout a joint's range of motion as a result of the change in mechanical advantage at various joint angles (2,4,8,12). In contrast, attaching chains to the bar theoretically provides a variable resistance so that when the barbell is lowered to the bottom of the movement, the chains pile up on the floor and decrease external resistance. When the bar is raised, more chain is hanging from the bar to increase the external resistance. Therefore, the rationale for using chains is that the external resistance more closely mimics the actions of body joints. This same concept is used with elastic band training, in which the band is attached to the bar and anchored to the floor. Several research studies have shown success with elastic bands and strength training (5,11,17–20,22). If chains, in fact, do work in the same manner as elastic bands, chains may provide an additional training stimulus not afforded by conventional weight alone.

Advocates of chain training suggest that exercises such as bench press, squats, and Olympic lifts (Snatch and Clean) increase strength, power, acceleration, motor control, stabilization, and enhanced neurologic adaptation more so than conventional barbell exercises alone (2,4,10,19,20). However, these claims remain mostly anecdotal, and to date, only 2 peer-reviewed studies have investigated the use of chains with traditional lifting techniques (4,6).

Ebben and Jensen (6) examined integrated electromyography for the quadriceps and hamstring muscle groups and ground reaction forces (GRFs) of 11 NCAA Division I athletes performing the back squat with traditional barbells,

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22(2)/390–395

Journal of Strength and Conditioning Research
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traditional barbells with chains accounting for 10% of the load, and traditional barbells with elastic bands attached to the ends of the bars. The results showed no significant differences between integrated electromyography and GRFs between traditional squats and squats using chains or elastic bands. However, the athletes stated that using chains and bands during squatting felt different, and this difference suggested that chains and bands may have some effect. Coker et al. (4) investigated the acute effects of chains on the Olympic Snatch on 7 competitive weightlifters who regularly used chains as part of their training program. Barbell vertical displacement, maximal velocity, rate of force development, and vertical GRFs compared the weightlifter's Olympic Snatch at 80% and 85% of 1 repetition maximum (1RM) using conventional barbells with 80% (75% conventional barbells + 5% chains) and 85% (80% conventional barbells + 5% chains) of 1RM using chains. No statistically significant differences were found to suggest that chains had no biomechanical effect. After lifting, the subjects completed a 2-item questionnaire regarding their perception of the use of chains. All subjects stated that they perceived that the chains forced them to pull harder throughout the lift. Additionally, all subjects felt that the chain oscillation in the catch phase of the Olympic Snatch required greater stabilization of their shoulders, abdominals, and back, enhancing their strength during training.

Anecdotal and perceptual information suggests that chains provide positive effects, but empirical research does not support this notion. Although perception is not always reality, chain training continues to grow rapidly and thus warrants further empirical research. This study hypothesized that the addition of chains during the Olympic Clean would require the lifter to pull the bar higher, faster, and with greater force compared to the same weight without chains. Therefore, the purpose of this study was to determine whether mechanical changes would result from the addition of chains during the execution of the Olympic Clean. In addition, follow-up questions evaluating the subjects' perception of the effect of chains was also investigated.

METHODS

Experimental Approach to the Problem

This study was designed to investigate the performance of the Olympic Clean with and without chains at the same given percentage of 1RM. The subjects were 7 (4 men and 3 women) trained and competitive weightlifters who currently used chains as part of their competitive training program. The subjects were initially tested for their 1RM in the Olympic Clean 7 to 10 days before data collection. During testing, each subject performed 5 Olympic Cleans. The testing order mimicked an actual weightlifting competition so that lifters would not lift in succession, and it allowed 5 to 8 minutes of recovery between sets. The first of the 5 lifts was performed at 75% of the subjects' 1RM to allow lifters the opportunity to become familiar with the lifting platform and

data collection procedure. The subjects then lifted 80% of their 1RM with conventional weights only and 80% (75% conventional weights + 5% chains) of their 1RM with conventional weights and chains. The subjects then followed the same procedure at 85% of their 1RM without chains and 85% (80% conventional weights + 5% chains) with conventional weights and chains. Force plate and video analysis were used to determine maximal vertical displacement of the bar, maximal bar velocity, rate of force production of the bar, and the vertical GRFs for the first-pull, unweighting, and second-pull phases of the lift. Each variable was examined by using a condition (bar vs. chain)-by-load (80 vs. 85%) analysis of variance with repeated measures on both factors ($P < 0.05$). After data acquisition, the subjects were asked to provide a written response to 2 questions regarding their perceptions of how the chains affected their performance in the Olympic Clean.

Subjects

The sample group consisted of 7 regionally and nationally competitive weightlifters (4 men and 3 women) with an average age of 31.0 ± 11.8 years, an average weight of 79.3 ± 17.9 kg, and an average height of 174.8 ± 0.1 cm, who were currently and had been competing between 1 and 13 years. Each weightlifter had held a state title in the Olympic Clean and Jerk, and several had held national records in their age classes during their lifting career. Every weightlifter used chains as part of their regular training program and in preparation for competitions. Subject descriptive data are reported in Table 1.

Weightlifting is a relatively small sport, so finding highly trained competitive individuals who regularly use chains is difficult. In turn, the sample size is small, and therefore, the authors recognize the size may affect the study statistically. However, chain training is rapidly growing among athletes and strength and conditioning coaches at the high school and collegiate levels. Unfortunately, most of what is known about chain training is anecdotal, and little empirical research exists to substantiate proposed benefits attributed to chain training. Therefore, the authors felt that these data needed to be disseminated to the strength and conditioning community.

TABLE 1. Descriptive data of the seven subjects.

	Mean $\pm$ SD	Range
Age (y)	31.0 ± 11.8	17.0–47.0
Height (cm)	174.8 ± 0.1	165.0–185.0
Weight (kg)	79.3 ± 17.9	57.5–104.9
1 repetition maximum		
Power Clean (kg)	86.4 ± 39.1	40.0–135.0
Years spent training with chains	5.9 ± 4.7	1.0–13.0

Approval for this study was provided by the university's institutional review board, and each subject completed a written informed consent form before experimental testing.

Procedures

The subjects were tested for their 1RM by their coach 7 to 10 days before data collection by using 1RM testing protocol established by Kraemer and Fry (14). From this result, 80% and 85% loads were determined. To determine appropriate chain loads, procedures outlined by Berning et al. (2) were used. First, various individual chain links (e.g., 1/4, 3/8, and 1/2 inch) were weighed. Because each lifter stood at a different height, individual length measurements were taken from the floor to the bar at the catch phase. By using the lifters' 80% and 85% of 1RM values, the chain length measurement, the fact that 1 chain is hanging off each side of the bar, and the weight of the individual chain links and chain sizes, a 5% chain load for each individual was computed.

When the subjects arrived at the testing facility, each completed a written consent to participate, and the testing procedures were explained. A warm-up area was provided, and each lifter used his or her own individual warm-up protocol under the supervision of his or her coach. The women completed the study first and were followed by the men. The subjects lifted in a sequential order based on weight class, as would be performed in competition. In this manner, all subjects completed their lift at a given percentage (e.g., 80% without chains) before moving to the next lift (e.g., 80% with chains). Doing so ensured that no subject would perform 2 lifts in succession, and thus, a minimum of 5 to 8 minutes of rest occurred between lifts. The subjects performed 5 trials consisting of 1 repetition each. The first trial was performed at 75% of their 1RM with conventional weights only. This trial allowed the subjects to become familiar with the laboratory, lifting platform, force plate, and data collection protocol. The remaining 4 trials occurred in the following sequence: 80% of the individual's 1RM using conventional weights only, 80% of the individual's 1RM with chains (75% conventional weights and 5% chains), 85% of the individual's 1RM using conventional weights only, and 85% of the individual's 1RM (80% conventional weight and 5% chains).

Individual Perceptions

Lifters were asked to provide written responses to the following questions after completing the study: Did you perceive a difference between performing Olympic Cleans with chains vs. without chains when the weight was the same (e.g., 80% of maximum without chains compared with 80% maximum with chains)? If yes, describe what you experienced (e.g., harder, easier, or whatever you felt).

Data Acquisition

During each trial, the subjects positioned their right foot (20) on an AMTI force plate (63.5 × 63.5 cm) (Advanced Medical Technologies, Inc., Watertown, MA), which collected GRFs at a rate of 1000 Hz. In addition, all trials were videotaped by using an S-VHS video camera (Panasonic AG-456; Panasonic Corp., Secaucus, NJ) capturing the subject's right side in the sagittal plane at 60 Hz. Kinematic data were digitized in accordance with the protocol suggested by Garhammer (9) and analyzed using the Peak Motus 4.3 motion analysis system (Peak Performance Technologies, Inc., Centennial, CO). Kinematic and kinetic data were synchronized by using the Peak Motus Event and Video Control Unit.

Statistical Analyses

All data were collected in a single testing session. The variables examined for each Olympic Clean condition (conventional vs. chains) were maximal vertical displacement of the bar, maximal bar velocity, rate of force development of the bar, and the vertical GRFs for the first-pull, unweighting, and second-pull phases of the Olympic Clean. Each variable was analyzed with a repeated-measures analysis of variance. By assuming an effect size of 1.5 *SDs* as noteworthy, 70% power can be approached ($\alpha = 0.05$) with 7 participants. Alpha levels were set at $P \leq 0.05$.

RESULTS

Means and *SDs* for all variables examined are provided in Tables 2 (80% load) and 3 (85% load). Results indicated that the rate of force development was significantly different for load ($F[1,12] = 35.83$; $P < 0.05$), whereby the subjects generated more power at the 85% load (mean = 1311.93 W; *SD* = 163.07) than for the 80% load (mean = 1222.42 W;

TABLE 2. Means and *SDs* for the Power Clean at eighty percent of one repetition maximum with and without chains.

Measured variables	Without chains	With chains
Bar displacement (m)	1.38 ± 0.05	1.37 ± 0.05
Bar velocity (m·s ⁻¹)	1.68 ± 0.15	1.70 ± 0.12
Ground reaction force for first pull (N)	885.00 ± 295.27	888.29 ± 271.38
Ground reaction force for unweighting (N)	449.90 ± 176.22	515.22 ± 194.39
Ground reaction force for second pull (N)	1,053.04 ± 282.68	1,136.08 ± 297.17
Rate of force development of bar (W)	1,228.73 ± 73.24	1,216.11 ± 100.27

TABLE 3. Means and SDs for the Power Clean at 85% of one repetition maximum with and without chains.

Measured variables	Without chains	With chains
Bar displacement (m)	1.38 ± 0.05	1.37 ± 0.05
Bar velocity (m·s ⁻¹)	1.67 ± 0.14	1.62 ± 0.10
Ground reaction force for first pull (N)	917.46 ± 311.24	883.36 ± 276.96
Ground reaction force for unweighting (N)	495.09 ± 251.25	490.57 ± 197.80
Ground reaction force for second pull (N)	1,128.37 ± 300.52	1,122.48 ± 307.45
Rate of force development of bar (W)	1,350.42 ± 186.44	1,273.44 ± 139.28

$SD = 84.61$). Furthermore, a significant condition-by-load interaction was found for the GRF of the second pull ($F[1,12] = 5.56$; $P < 0.05$). A Tukey post hoc analysis showed that the GRF of the second pull was significantly greater for the 85% (mean = 1,128.36 N; $SD = 300.52$) than for the 80% load (mean = 1,053.04 N; $SD = 282.76$) for the bar only condition. Simple main effects for condition were not statistically significant.

Responses to the questionnaire indicated that 100% of the subjects stated that there was a definite effect associated with using chains. When they compared the Power Clean with and without chains, the subjects suggested that using chains was more difficult. The subjects also stated that they felt they had to pull harder to complete the lift because of the added weight of the chains throughout the lift.

DISCUSSION

The purpose of this study was to determine whether mechanical changes would result from the addition of chains during the execution of the Olympic Clean. Additionally, follow-up questions evaluating the subjects' perception of the effect of chains were also investigated. Anecdotal claims suggest that the addition of chains to traditional exercises, such as the squat, bench press, and Olympic lifts, requires greater effort, in turn leading to greater strength, joint stability, power output, motor control, and neuromuscular adaptations (4,10,19,20). Therefore, it was hypothesized that the addition of chains during the Olympic Clean would require the lifter to pull the bar higher, faster, and with greater force, exhibiting greater GRFs compared to the same weight without chains. However, the results of this study do not support this hypothesis. These findings were similar to those of Ebben and Jensen (6) and Coker et al. (4), supporting the idea that chains do not elicit any mechanical difference compared to lifting weight without chains, if the load remains constant.

Several factors may have contributed to the results of this study. The sample size and statistical power were low. Similarly, the studies by Ebben and Jensen (6) and Coker et al. (4) had small sample sizes, and both studies indicated that chains had no effect on performance. This finding suggests that

larger sample sizes may be necessary to determine whether chains have a valid effect on performance.

An alternative explanation may be that the percentage of the load that was accounted for by the chains was not great enough to elicit an effect. In contrast, Wallace et al. (22) found significant differences with greater load percentages when investigating the effects of elastic bands on peak force, peak power, and peak rate of force development during the back squat exercise on 10 (4 women and 6 men) recreationally resistance-trained subjects. After determining 1RM in the back squat, the subjects performed 2 sets of 3 repetitions at 60% and 85% of 1RM, with and without elastic bands, on 2 separate days. Although the loads were established at 60% and 85% of 1RM, the authors further divided the study into 3 conditions: no bands (NB), B1, and B2 groups. NB conditions required lifters to perform back squats using conventional weights only at 60% and 85% of 1RM. B1 and B2 conditions required lifters to perform back squats using a combination of conventional weights and bands, with B1 providing 20% of the 1RM load and B2 providing 35% of the 1RM load. The results indicated that the subjects exerted significantly greater mean force values and mean power values when the bands accounted for 20% or 35% of the load, but only when the subjects were squatting 85% of their 1RM as compared to squatting 85% of their 1RM without bands. No significant differences occurred under any condition at 60% of the subjects' 1RM.

In the current study, chains accounted for only 5% of the total load. Likewise, Coker et al. (4) used only 5% of the load with chains and Ebben and Jensen (6) used 10% of the load with chains. The lift percentages used in the current study were similar (80 vs. 85% of 1RM) to those used in the band study. Because of the significant findings of Wallace et al. (21), using a higher chain load percentage (e.g., 20% or more) may be necessary to achieve similar results.

The second part of this study investigated individual lifter's perceptions regarding the effect of chains on their performance in the Olympic Clean. Ebben and Jensen (6) and Coker et al. (4) asked similar questions of their subjects at the conclusions of their studies. Ebben and Jensen (6) asked their subjects whether they felt a difference between performing

the squat exercise with and without chains. Their subjects reported that the 2 methods felt different, but there was no elaboration on how they felt the squat was different with and without chains.

Similarly, Coker et al. (4) asked their subjects 2 questions. The first question asked: Did you perceive a difference between performing the snatch with chains vs. without chains when the weights were the same (e.g., 80% of maximum conventional weight vs. 80% of maximum with chains)? The follow-up question asked: If yes, describe what you experienced (e.g., harder or easier). All subjects responded that they felt a difference between the 2 lifts (chains vs. no chains). The subjects indicated that they felt they had to apply a greater amount of force when performing the Olympic Snatch with chains and that greater strength was needed to stabilize the bar in the catch position. Additionally, the subjects emphasized that training with chains as part of their precompetition program unquestionably allowed them to control heavier weights and, in turn, lift more weight during competition. This finding provided a strong argument for the use of chains in training. It should be recalled, however, that although lifters perceived a difference, statistical analysis indicated that there was no difference biomechanically, emphasizing that perception is not always reality.

To follow the protocol established by these 2 studies, lifters in the current study were asked 2 similar questions posed by Coker et al. (4). The first question addressed individual perception and asked: Did you perceive a difference between performing the Olympic Clean with chains vs. without chains at the same given weight load (e.g., 80% with chains vs. 80% without chains)? A follow-up question asked: If yes, can you describe what you perceived to experience? Similar to the subjects in the studies by Ebben and Jensen (6) and Coker et al. (4), all lifters perceived that chains affected their lifting performance. All lifters commented that the chains caused them to have to exert a greater force and pull harder. Additionally, the lifters rationalized that repeated use of chains as part of their regular training program provided them with the ability to control heavier weights in competition. The lifters' perceptions in these 3 studies may explain in part why chain training is gaining support and popularity and may provide some insight as to the rapid popularity growth at all levels of sport and competition despite a lack of empirical data supporting its use.

The results from this and past research (4–6,17,22) on the use of chains and bands are mixed. One possible reason may simply be that chains do not require greater physiologic effects but are only perceived to do so. In contrast, perhaps the weight loads used in previous research were not great enough to elicit a physiologic response, or perhaps physiologic effects depend on specific exercises at specific percentages of the maximum. Another possible explanation may involve sample size. Larger sample sizes may be necessary, but larger sizes potentially pose a problem for weightlifting

in that finding competitive and highly trained weightlifters who regularly use chains as part of their training may be difficult.

Another potential explanation for the nonsignificant results may lie in the fact that the lifters were competitive and trained regularly with chains. Neuromuscular and technical adaptations may have already occurred such that kinematic and kinetic differences were slight and not statistically significant. According to Ericsson (7), "Expert performers attain the ability to consistently reproduce the same motor actions." This notion was recently supported by Jeansson et al. (15), who found that experts did not use a significantly different lift pattern for the Clean and Jerk under 2 different weight conditions, 49.55 kg (30–40% of 1RM) and 70.85 ± 11 kg (60% of 1RM). However, Jeansson et al. (15) found this to be true at lower percentages of 1RM and possibly would have found differences if variables were measured at higher percentages of 1RM, as found in this study at 80% and 85% of 1RM. This may imply that there is a point at which physiologic differences begin to occur and a given percentage of 1RM at which chains start to affect performance.

As a result of this study and others, 4 main areas need further investigation. First, future studies must investigate the use of chains on highly trained and novice populations. This study used trained lifters already experienced with chains, whereas the possibility exists that novice lifters may benefit from incorporating chains in their training. However, caution should be used because novice lifters may experience unwanted changes in lifting form as a result of the forces imposed by the chains. Second, this study investigated performance at only 80% (75% conventional weights + 5% chain) and 85% (80% conventional weights + 5% chain) of lifters' Olympic Clean 1RM. Future investigations should consider conventional loads over an entire spectrum of 1RM (e.g., 10–100% of 1RM). Third, research must investigate what the optimal percentage or percentage range of the chain load should be (e.g., 5%, 10%, or 20%). This study used 5% of the lifters' 1RM, whereas others used 10% (6). Potentially, another percentage may prove more advantageous, such as the 20% of load used by Wallace et al. (22) in their band study. Lastly, all chain studies to date have evaluated the acute effects of chains on the squat or Olympic lifts. The need for training studies must occur if athletes and strength coaches want to implement chains effectively into a regular training program and elicit positive effects. The notion that chains work is currently supported by many athletes and strength coaches alike but is supported based on only popularity and anecdotal information.

In conclusion, anecdotal evidence remains strong, and individual perceptions that chains positively increase strength, power, and performance continue to grow. However, empirical data do not support these notions. This study did not support the hypothesis that the use of chains during the Olympic Clean required greater effort than performing the same lift at the same loads without chains. In contrast, lifters'

subjective responses indicated that chains, in fact, make the lifting task more difficult and required a greater effort to complete the lifts effectively. To bridge the gap between empirical and perceptual data, further investigation is required, especially in training studies. Until then, athletes and strength coaches will continue to use chains in their training, and if athletes and strength coaches perceive that chains elicit a positive effect and no adverse effects come from chain use, perhaps the psychological aspect will allow individuals to train harder and ultimately result in greater force, power, and performance.

PRACTICAL APPLICATIONS

The use of chains in commercial and competitive environments as an addition to conventional training continues to grow. A pro-con argument as to the practicality of implementing chains exists. The perceptual results of this study and others (4,6) and anecdotal support from advocates of chain training (3,10,11,13,16,19–21) support its use as an addition to conventional lifting. Advocates claim that chains positively affect their training, require greater efforts, and, in turn, increase performance. Practically, if an individual perceives chain training as more difficult and it allows him or her to work harder during regular training, greater exertion may lead to increased performance over time. If the use of chains does not pose a danger to the lifter, perhaps their use is warranted. However, because of the complexities of determining appropriate chain weight percentages of 1RMs, creating routines, and establishing appropriate sets, repetitions, and volume, this technique may be advisable for only individual competitors, such as powerlifters, rather than for team sports, in which training large numbers of athletes occurs.

Although using chains is anecdotally and perceptually supported, it must be considered that the results of empirical research indicate that there is no benefit to using chains. Based on current studies, an athlete would gain as much, if not more, from simply adding another plate rather than from adding chains. Additionally, the complexity associated with chains (e.g., determining chain weight, calculating lifter heights, and computing percentages of 1RM) requires more time than the potential value gained from including chains. Because no data exist as to the ideal percentage necessary to maximize potential, implementation is haphazard at best. Practically, most strength and conditioning coaches have a limited time to train athletes, and the use of conventional weights without chains is simpler, more efficient, and more practical. Additionally, no practical suggestions can be made regarding appropriate sets, repetitions, percentages of 1RM, or chain training routines, and therefore, those who use chains do so by using a best guess approach. Therefore, the authors recommend that chain training should be used only as a supplemental technique to conventional weight training and do not recommend chains as a mainstay for training.

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